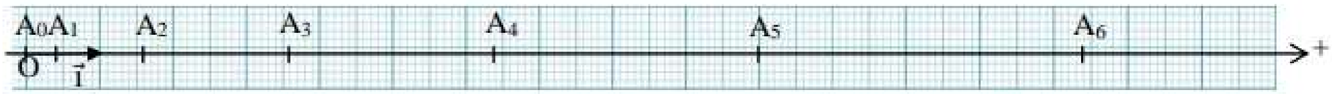
The following recording shows, to real scale, the positions occupied by a mobile (A) separated by equal time intervals τ = 40 ms. On the date $t_0 = 0$, (A) is at A_0.



1. The movement of the mobile is rectilinear. Justify.
2. Calculate the average speed of the mobile (A) between the dates: t_0 and t_4 ; t_2 and t_5.
3. Calculate the instantaneous speed of the moving body (A) at the instants t_1, t_2, t_3, t_4, and t_5.
4. What can we conclude about the type of motion?
5. Represent, on the same graph, the instantaneous velocity vector $\vec{v}_3$ at the moment t_3, using the scale: 1 cm → 0.2 m/s
6. Calculate the average acceleration between the instants: t_1 and t_5.
7. Calculate the instantaneous acceleration at the dates: t_2 and t_4 respectively. Conclude.
8. Sketch, on the same graph, the acceleration vector $\vec{a}_4$ using the scale: 1 cm → 2 m/s^2.